

# The Hellings-Downs curve for a Gaussian source

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This paper is devoted to the derivation of the Hellings-Downs curve for the case of a Gaussian source of the Gravitational Wave Background.

## 1 Problem Statement

Consider the spherical coordinate system  $(\phi, \theta, r)$ , where  $\phi$  is longitude,  $\theta$  is colatitude and  $r$  is radius. Then the orthonormal basis in the cartesian coordinate system will look like:

$$\hat{\Omega} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad (1a)$$

$$\hat{n} = (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta), \quad (1b)$$

$$\hat{m} = (\sin \phi, -\cos \phi, 0), \quad (1c)$$

where  $(\hat{\Omega}, \hat{n}, \hat{m}) \equiv (\hat{r}, \hat{\theta}, -\hat{\phi})$ . Since  $\bar{r} = r \cdot \hat{r}$ , and the point on the unit sphere has the value  $r = 1$ , we can describe the unit sphere with a single vector  $\hat{\Omega} = \hat{r}$ .

We also know that gravitational waves have two polarization, which are called "plus" and "cross", and are described by the corresponding tensors:

$$e_{\alpha\beta}^+(\hat{\Omega}) = \hat{m}_\alpha \hat{m}_\beta - \hat{n}_\alpha \hat{n}_\beta, \quad (2a)$$

$$e_{\alpha\beta}^\times(\hat{\Omega}) = \hat{m}_\alpha \hat{n}_\beta + \hat{n}_\alpha \hat{m}_\beta \quad (2b)$$

Then, frequency shift of pulsar radiation  $z = \Delta\nu/\nu_0$  in Fourier space:

$$\tilde{z}(f, \hat{\Omega}) = \left( e^{-i2\pi f L(1+\hat{\Omega} \cdot \hat{p})} - 1 \right) h(f, \hat{\Omega}) F_{\hat{p}}(f, \hat{\Omega}), \quad (3)$$

where  $\hat{p}$  is the unit vector in the direction of the pulsar,  $L$  is the distance to the pulsar,  $h(f, \hat{\Omega})$  is spectrum of gravitational waves and  $F_{\hat{p}}(\hat{\Omega})$  is pulsar response:

$$F_{\hat{p}}(\hat{\Omega}) = e_{\alpha\beta}(\hat{\Omega}) \frac{1}{2} \frac{\hat{p}^\alpha \hat{p}^\beta}{1 + \hat{\Omega} \cdot \hat{p}}. \quad (4)$$

Here we use a complex representation of the polarization tensor and spectrum, and in the case where the GWB is unpolarized, we have:

$$h(f, \hat{\Omega}) = h_+(f, \hat{\Omega}) + i \cdot h_\times(f, \hat{\Omega}) \quad (5)$$

$$e_{\alpha\beta}(\hat{\Omega}) = e_{\alpha\beta}^+(\hat{\Omega}) + i \cdot e_{\alpha\beta}^\times(\hat{\Omega}) \quad (6)$$

Since the Hellings-Downs curve relates correlations between signals from two pulsars, we defined the

Hellings-Downs curve as:

$$\Gamma[\mathcal{P}](\hat{p}, \hat{q}) = \frac{3}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot \mathcal{P}(\hat{\Omega}), \quad (7)$$

where  $\mathcal{P}(\hat{\Omega})$  is the power spectrum on the unit sphere, and with the frequency power spectrum  $\mathcal{H}(f)$  defined as:

$$|h(f, \hat{\Omega})|^2 = \mathcal{H}(f) \cdot \mathcal{P}(\hat{\Omega}), \quad (8)$$

and  $\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q})$  is the integral kernel, which is defined as:

$$\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) = \mathcal{R} \left[ F_{\hat{p}}^*(\hat{\Omega}) \cdot F_{\hat{q}}(\hat{\Omega}) \right]. \quad (9)$$

And in our case, we assume that the power spectrum is Gaussian and looks like this:

$$\mathcal{P}(\hat{\Omega}) = \exp \left[ -\frac{(\hat{\Omega} - \hat{\Omega}_0)^2}{2\sigma^2} \right], \quad (10)$$

where  $\hat{\Omega}_0$  is the center of the source and  $\sigma$  is the width of the Gaussian. Converting the numerator:

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = \hat{\Omega}^2 + \hat{\Omega}_0^2 - 2\hat{\Omega} \cdot \hat{\Omega}_0 = 1 + 1 - 2\hat{\Omega} \cdot \hat{\Omega}_0$$

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = 2(1 - \hat{\Omega} \cdot \hat{\Omega}_0),$$

we have:

$$\mathcal{P}(\hat{\Omega}) = \exp \left[ -\frac{2(1 - \hat{\Omega} \cdot \hat{\Omega}_0)}{2\sigma^2} \right],$$

$$\mathcal{P}(\hat{\Omega}) = \exp \left[ -\frac{1}{\sigma^2} \right] \cdot \exp \left[ \frac{\hat{\Omega} \cdot \hat{\Omega}_0}{\sigma^2} \right].$$

Let  $\kappa = 1/\sigma^2$ , and as a result we have:

$$\mathcal{P}(\hat{\Omega}) = \exp(-\kappa) \cdot \exp(\kappa \hat{\Omega} \cdot \hat{\Omega}_0). \quad (11)$$

And to find the Hellings-Downs curve for a Gaussian source, we need to calculate:

$$\Gamma^{(g)}(\hat{p}, \hat{q}) = \frac{3e^{-\kappa}}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (12)$$

## 2 Integral in Coordinates

Firstly, let's represent the integral kernel  $\mathcal{K}$  in vector form:

$$\mathcal{K} = \mathcal{R} \left[ e_{\alpha\beta}^*(\hat{\Omega}) \frac{1}{2} \frac{\hat{p}^\alpha \hat{p}^\beta}{1 + \hat{\Omega} \cdot \hat{p}} \cdot e_{\mu\nu}(\hat{\Omega}) \frac{1}{2} \frac{\hat{q}^\mu \hat{q}^\nu}{1 + \hat{\Omega} \cdot \hat{q}} \right],$$

$$\mathcal{K} = \frac{1}{4} \frac{\hat{p}^\alpha \hat{p}^\beta \hat{q}^\mu \hat{q}^\nu}{(1 + \hat{\Omega} \cdot \hat{p})(1 + \hat{\Omega} \cdot \hat{q})} \mathcal{R} \left[ e_{\alpha\beta}^*(\hat{\Omega}) \cdot e_{\mu\nu}(\hat{\Omega}) \right],$$

where:

$$\mathcal{R} \left[ e_{\alpha\beta}^*(\hat{\Omega}) \cdot e_{\mu\nu}(\hat{\Omega}) \right] = e_{\alpha\beta}^+(\hat{\Omega}) e_{\mu\nu}^+(\hat{\Omega}) + e_{\alpha\beta}^\times(\hat{\Omega}) e_{\mu\nu}^\times(\hat{\Omega}),$$

then:

$$\mathcal{K} = \frac{1}{4} \frac{\hat{p}^\alpha \hat{p}^\beta \hat{q}^\mu \hat{q}^\nu \left[ e_{\alpha\beta}^+(\hat{\Omega}) e_{\mu\nu}^+(\hat{\Omega}) + e_{\alpha\beta}^\times(\hat{\Omega}) e_{\mu\nu}^\times(\hat{\Omega}) \right]}{(1 + \hat{\Omega} \cdot \hat{p})(1 + \hat{\Omega} \cdot \hat{q})}. \quad (13)$$

Consider individual tensor convolutions:

$$\hat{v}^\alpha \hat{v}^\beta e_{\alpha\beta}^+(\hat{\Omega}) = (\hat{v} \cdot \hat{m})^2 - (\hat{v} \cdot \hat{n})^2, \quad (14a)$$

$$\hat{v}^\alpha \hat{v}^\beta e_{\alpha\beta}^\times(\hat{\Omega}) = 2(\hat{v} \cdot \hat{m})(\hat{v} \cdot \hat{n}). \quad (14b)$$

Let's denote  $P_a = (\hat{p} \cdot \hat{a})$  and  $Q_a = (\hat{q} \cdot \hat{a})$ , then:

$$\mathcal{K} = \frac{1}{4} \frac{(P_m^2 - P_n^2) \cdot (Q_m^2 - Q_n^2) + 4P_m P_n Q_m Q_n}{(1 + P_\Omega)(1 + Q_\Omega)}. \quad (15)$$

Note that since  $(\hat{m}, \hat{n}, \hat{\Omega})$  is an orthonormal basis, and  $|\hat{p}| = 1$ ,  $|\hat{q}| = 1$ , it means that:

$$P_n^2 = 1 - P_m^2 - P_\Omega^2,$$

$$Q_n^2 = 1 - Q_m^2 - Q_\Omega^2.$$

Then:

$$\frac{(P_n^2 - P_m^2)}{1 + P_\Omega} = \frac{1 - 2P_m^2 - P_\Omega^2}{1 + P_\Omega} = (1 - P_\Omega) - \frac{2P_m^2}{1 + P_\Omega},$$

$$\frac{(Q_n^2 - Q_m^2)}{1 + Q_\Omega} = \frac{1 - 2Q_m^2 - Q_\Omega^2}{1 + Q_\Omega} = (1 - Q_\Omega) - \frac{2Q_m^2}{1 + Q_\Omega}.$$

As a result, we can divide the integral kernel into four parts:

$$\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) = \frac{1}{4} (\mathcal{K}_I + \mathcal{K}_J + \mathcal{K}_K + \mathcal{K}_H), \quad (16)$$

where:

$$\mathcal{K}_I = (1 - P_\Omega)(1 - Q_\Omega), \quad (17a)$$

$$\mathcal{K}_J = -2 \cdot \frac{P_m^2(1 - Q_\Omega)}{(1 + P_\Omega)}, \quad (17b)$$

$$\mathcal{K}_K = -2 \cdot \frac{Q_m^2(1 - P_\Omega)}{(1 + Q_\Omega)}, \quad (17c)$$

$$\mathcal{K}_H = 4 \cdot \frac{P_m Q_m (P_m Q_m + P_n Q_n)}{(1 + P_\Omega)(1 + Q_\Omega)}, \quad (17d)$$

and we can divide the integral into the same four parts:

$$\Gamma^{(g)}(\hat{p}, \hat{q}) = \frac{3e^{-\kappa}}{4\pi} \cdot \frac{1}{4} (I + J + K + H), \quad (18)$$

where:

$$I = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_I \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (19a)$$

$$J = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_J \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (19b)$$

$$K = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_K \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (19c)$$

$$H = \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}_H \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (19d)$$

Let's define a convenient coordinate system. As we can see, all integrals include a term containing  $(\hat{\Omega} \cdot \hat{\Omega}_0)$  in the exponent, and to simplify this term, we take  $\hat{\Omega}_0 \equiv (0, 0, 1)$ , because then  $(\hat{\Omega} \cdot \hat{\Omega}_0) = \cos \theta$ . For the pulsar vectors, we have:

$$\hat{p} = (\sin \alpha, 0, \cos \alpha) \quad (20a)$$

$$\hat{q} = (\sin \beta \cos \gamma, \sin \beta \sin \gamma, \cos \beta) \quad (20b)$$

where  $\alpha, \beta$  are the angles between the source center and the pulsars, and  $\gamma$  is the angle between the directions to the pulsars from the source center. Now we can calculate all the scalar products:

$$P_\Omega = \sin \alpha \sin \theta \cos \phi + \cos \alpha \cos \theta,$$

$$Q_\Omega = \sin \beta \sin \theta \cos(\phi - \gamma) + \cos \beta \cos \theta,$$

$$P_n = \sin \alpha \cos \theta \cos \phi - \cos \alpha \sin \theta,$$

$$Q_n = \sin \beta \cos \theta \cos(\phi - \gamma) - \cos \beta \sin \theta$$

$$P_m = \sin \alpha \sin \phi,$$

$$Q_m = \sin \beta \sin(\phi - \gamma);$$

Also, let's highlight the dependence of the coefficients on  $\phi$ :

$$P_\Omega = a \cos \phi + b, Q_\Omega = c \cos(\phi - \gamma) + d, \quad (21a)$$

$$P_n = h \cos \phi - s, Q_n = k \cos(\phi - \gamma) - t, \quad (21b)$$

$$P_m = f \sin \phi, Q_m = g \sin(\phi - \gamma); \quad (21c)$$

where:

$$a = \sin \alpha \sin \theta, b = \cos \alpha \cos \theta, \quad (22a)$$

$$c = \sin \beta \sin \theta, d = \cos \beta \cos \theta, \quad (22b)$$

$$h = \sin \alpha \cos \theta, s = \cos \alpha \sin \theta, \quad (22c)$$

$$k = \sin \alpha \cos \theta, t = \cos \alpha \sin \theta, \quad (22d)$$

$$f = \sin \alpha, g = \sin \beta, \quad (22e)$$